

Dilusha Chandrasiri

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About

Computer Science and Engineering undergraduate at the University of Moratuwa, specializing in Data Science and Engineering. Passionate about building intelligent systems through machine learning, data analytics, and scalable software engineering. Seeking an internship to further strengthen technical, systems design, and product development skills in a collaborative environment.

Education

University of Moratuwa

BSc (Hons) in Computer Science and Engineering

- CGPA: 3.52 / 4.00 | Specializing in Data Science and Engineering

2024 – Present

Colombo, Sri Lanka

Maliyadeva College, Kurunegala

G.C.E. Advanced Level – Physical Science Stream

- 3As | Z-Score: 2.4649 | District Rank: 20 | Island Rank: 266

2023

Kurunegala, Sri Lanka

Experience

Chrisentia Solutions

AI & Computer Vision Intern

- Developed a YOLO-based coconut detection and counting system for Wicky Plantation Company (Pvt) Ltd using computer vision on an enclosed conveyor tunnel setup.
- Contributed to the full pipeline including data collection, annotation, model training, evaluation, and deployment for agricultural automation workflows.

April 2026 – Present

Sri Lanka

LLM Task Force

Volunteer Research Contributor

- Contributing to IEEE LLM Task Force research on Large Language Models, sovereign AI, and government sector AI applications.
- Improving a modular multi-agent RAG system using LangGraph for collaborative reasoning, structured retrieval, and agent-based query decomposition.

March 2026 – Present

International

Projects

Personal Projects

Xyra – WhatsApp AI Agent for Micro Businesses 🔄

Ongoing

Next.js, FastAPI, LangGraph, OpenAI, PostgreSQL, pgvector, Celery, Redis, Docker

- Building a production-grade multi-tenant AI agent for Sri Lankan micro businesses to handle customer orders, FAQs, and appointment bookings in Sinhala, Tamil, and English via WhatsApp and web chat.
- Architected a LangGraph state machine with tool-calling, hybrid RAG (keyword + pgvector), and a document ingestion pipeline with Redis for async processing.
- Designed a channel-agnostic message router enabling seamless web chat simulator to WhatsApp Cloud API migration with no changes to agent, DB, or dashboard.

DSA Genie – AI-Powered DSA Chatbot 🔄

April 2026

React, FastAPI, LangChain, FAISS, Ollama, HuggingFace, Python

- Built a DSA chatbot using LangChain, FAISS vector search, and HuggingFace sentence-transformer embeddings over a DSA PDF knowledge base with llama3.2 via Ollama.

- Implemented SSE-based streaming responses with structured output (explanation, algorithm, code, complexity analysis) and a sandboxed in-browser Python/JavaScript code runner.
- Exposed code execution tools via an MCP server using `fastmcp`, enabling integration with any MCP-compatible AI client.

TravelMate – AI Travel Recommendation Platform

March 2026

React, FastAPI, Python, PostgreSQL, scikit-learn, pandas, Vite

- Built an AI-powered travel companion for Sri Lanka with natural-language preference matching using TF-IDF and cosine similarity over traveler review data.
- Developed a FastAPI backend with endpoints for destination discovery, location browsing by city and type, and top review retrieval from PostgreSQL.
- Deployed frontend on Vercel and API on Render; seeded the database from a CSV dataset using a Node.js import script with `csv-parser`.

MoneyMap – AI-Powered Personal Finance Assistant

February 2026

React, Node.js, Express, PostgreSQL, Python, Flask, Tesseract OCR, JWT, Firebase

- Full-stack finance tracker with dashboard, analytics, and multi-category expense management with JWT & Firebase OAuth authentication.
- Built a Python Flask OCR microservice using Tesseract to extract store name, date, and totals from receipt images and auto-fill transactions.
- Architected a 3-service system (React frontend, Node.js API, Flask OCR) with PostgreSQL, real-time charts via Recharts, and a RESTful API layer.

DocuMind – Privacy-First Local RAG System

January 2026

Python, LangChain, Ollama, FAISS, Sentence Transformers, PyPDF, React, FastAPI

- Built a fully local RAG system for conversational Q&A over PDFs using Ollama LLM inference and FAISS vector search.
- Implemented a modular pipeline: PDF ingestion → text chunking → embedding → FAISS indexing → context-aware answer generation with page-level source references.
- Exposed a FastAPI backend with internal modules, paired with a React frontend for document upload and query interface.

Academic Projects

Intelligent Traffic System

May 2026

OpenCV, YOLO, PyTorch, Kafka, Flink, PostgreSQL, Next.js, Mapbox, Docker

- AI-powered traffic monitoring system using YOLO and PyTorch for real-time vehicle detection, classification, and congestion prediction from video streams.
- Designed a real-time stream processing pipeline with Kafka and Flink for high-throughput traffic metadata ingestion and processing.
- Built a Next.js dashboard with Mapbox heatmaps and live congestion visualization; fully containerized with Docker.

Clinic Appointment & Treatment Management System

October 2025

Next.js, React, PostgreSQL, REST API

- Full-stack clinic management system with patient records, appointment scheduling (conflict detection, emergency flags), treatments, invoicing, and insurance claims.
- Implemented role-based authentication (admin/staff), automated invoice generation via PostgreSQL triggers, and staff action logging.
- Built reporting endpoints for branch revenue, doctor performance, treatment analysis, and outstanding balances across multi-branch operations.

Skills

Programming Languages: Python, JavaScript, Java, C++

Web Technologies: FastAPI, React, Node.js, Express, Next.js

Database Systems: MongoDB, MySQL, PostgreSQL

Data Science & ML: PyTorch, OpenCV, LangGraph, LangChain, YOLO

Tools & Technologies: Docker, Git, Jupyter Notebook, Kafka

Languages: English (Fluent), Sinhala (Native)

Awards & Achievements

4th Place – ENIGMA 2.0, Competitive Programming Competition	2025
Finalist – SLLIT Xstream Hackathon	2025
Participant – IEEEExtreme Programming Competition	2025
Coloursman – Chess, University of Moratuwa	2024 & 2025

Research

Inside the AI Model That Understands Student Performance (*Unpublished*)

[View Paper](#) 

How Environment and Urbanization Shape Bird Diversity in Sri Lanka (*Unpublished*)

[View Paper](#) 

Volunteering & Extra-Curricular

Technical Lead – IEEE MERCon 2026, University of Moratuwa

Volunteer Management Co-Lead – SLIoT Challenge 2026

Program Committee Member – SLIoT Challenge 2026

Project Chair – IEEE CIS Mind Models 2026, Sri Lanka Section

Program Committee Member – Career Fair 2026, Dept. of CSE, University of Moratuwa

Program Committee Member – 7th Mora Kings International Chess Championship 2025

Technical Committee Member – IEEE MERCon 2025, University of Moratuwa

Certifications

Postman API Fundamentals – Student Expert Certification

100 Days of Code: The Complete Python Pro Bootcamp – Angela Yu, Udemy

The Complete Web Development Bootcamp – Angela Yu, Udemy

AWS Academy Cloud Foundations – Amazon Web Services

AWS Academy – Microservices & CI/CD – Amazon Web Services

Kaggle Certifications – Intro to Machine Learning, Pandas, Data Cleaning, Feature Engineering

References

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Chandrashekar Konda

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